Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 6

Lab Report Name : Functions in SQL (String, Numeric, Date/Manipulation)

Course Title : Database Management System Sessional

Course Code : CSE 314

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Level : 3 **Term** : I **Section** : A **Group** : 1

Date of Submission : 15/12/2025 **Semester** : **Fall** **Year** : 2025

Key Learnings :

- Learned how single-row functions work on each row individually, unlike aggregate functions.
- Understood different categories of functions: String, Numeric, and Date/Time.
- Practiced using string functions such as UPPER, LOWER, CONCAT, SUBSTRING, and LENGTH.
- Gained experience with numeric functions like ROUND, ABS, and MOD for calculations.
- Learned how date/time functions like NOW(), YEAR(), and MONTH() extract useful information from dates.

Code Implementation and Output:

Select XAMPP for Windows - mysql -u root -p

```
MariaDB [uni]> create table student(
    -> student_id INT PRIMARY KEY,
    -> name VARCHAR(200),
    -> AGE int,
    -> DOB date,
    -> dept VARCHAR(200));
Query OK, 0 rows affected (0.030 sec)

MariaDB [uni]> INSERT INTO student
    -> VALUES
    -> (1,'Jiangly',20,'2005-12-12', 'CSE'),
    -> (2,'Petr',19,'2000-12-11','CSE'),
    -> (3,'Errichto',21,'2000-12-10','EEE'),
    -> (4,'Benq ',23,'2007-12-11','BBA');
Query OK, 4 rows affected (0.004 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
MariaDB [uni]> select name, UPPER(name) AS uppercase, LOWER(name) as lowercase
    -> FROM student;
+----------+-----------+-----------+
| name     | uppercase | lowercase |
+----------+-----------+-----------+
| Jiangly  | JIANGLY   | jiangly   |
| Petr     | PETR      | petr      |
| Errichto | ERRICHTO  | errichto  |
| Benq     | BENQ      | benq      |
+----------+-----------+-----------+
4 rows in set (0.000 sec)

MariaDB [uni]> SELECT CONCAT(name, '-', dept) AS student_info
    -> FROM student;
+--------------+
| student_info |
+--------------+
| Jiangly-CSE  |
| Petr-CSE     |
| Errichto-EEE |
| Benq -BBA    |
```

```
MariaDB [uni]> SELECT name, SUBSTRING(name, 1, 5) AS short_name
    -> FROM student;
+----------+------------+
| name     | short_name |
+----------+------------+
| Jiangly  | Jiang      |
| Petr     | Petr       |
| Errichto | Erric      |
| Benq     | Benq       |
+----------+------------+
4 rows in set (0.000 sec)

MariaDB [uni]> SELECT name, LENGTH(name) AS name_length
    -> FROM student;
+----------+-------------+
| name     | name_length |
+----------+-------------+
| Jiangly  |           7 |
| Petr     |           4 |
| Errichto |           8 |
| Benq     |           5 |
+----------+-------------+
4 rows in set (0.000 sec)

MariaDB [uni]> SELECT name, ROUND(age/3, 2) AS divided_age
    -> FROM student;
+----------+-------------+
| name     | divided_age |
+----------+-------------+
| Jiangly  |        6.67 |
| Petr     |        6.33 |
| Errichto |        7.00 |
| Benq     |        7.67 |
+----------+-------------+
```

```
MariaDB [uni]> SELECT ABS(-25) AS age_value;
+-----------+
| age_value |
+-----------+
|        25 |
+-----------+
1 row in set (0.000 sec)

MariaDB [uni]> SELECT name, MOD(age, 2) AS age_parity
    -> FROM student;
+----------+------------+
| name     | age_parity |
+----------+------------+
| Jiangly  |          0 |
| Petr     |          1 |
| Errichto |          1 |
| Benq     |          1 |
+----------+------------+
4 rows in set (0.000 sec)

MariaDB [uni]> SELECT name, YEAR(dob) AS birth_year,
    -> MONTH(dob) AS birth_month
    -> FROM student;
+----------+------------+-------------+
| name     | birth_year | birth_month |
+----------+------------+-------------+
| Jiangly  |       2005 |          12 |
| Petr     |       2000 |          12 |
| Errichto |       2000 |          12 |
| Benq     |       2007 |          12 |
+----------+------------+-------------+
4 rows in set (0.000 sec)
```

```
MariaDB [uni]> SELECT name,TIMESTAMPDIFF(YEAR, dob, CURDATE()) AS calculated_age
    -> FROM student;
+----------+----------------+
| name     | calculated_age |
+----------+----------------+
| Jiangly  |             19 |
| Petr     |             24 |
| Errichto |             24 |
| Benq     |             17 |
+----------+----------------+
4 rows in set (0.000 sec)

MariaDB [uni]> SELECT name, DATE_ADD(dob, INTERVAL 18 YEAR) AS adult_date
    -> FROM student;
+----------+------------+
| name     | adult_date |
+----------+------------+
| Jiangly  | 2023-12-12 |
| Petr     | 2018-12-11 |
| Errichto | 2018-12-10 |
| Benq     | 2025-12-11 |
+----------+------------+
4 rows in set (0.000 sec)

MariaDB [uni]> SELECT NOW() AS currentTime;
+---------------------+
| currentTime         |
+---------------------+
| 2025-12-07 22:57:54 |
+---------------------+
1 row in set (0.000 sec)
```